What happens when a patient’s breathing begins to deteriorate — but the change is not recognised early enough?

In COPD, the weeks between two consultations are usually the least visible part of the disease.

THE PATH FROM A BAD DAY TO A HOSPITAL VISIT

01 EVERYDAY BREATHING

Good days and bad days, managed at home

→

02 WORSENING SYMPTOMS

A change the patient feels — rarely written down

→

03 HEALTHCARE INTERVENTION

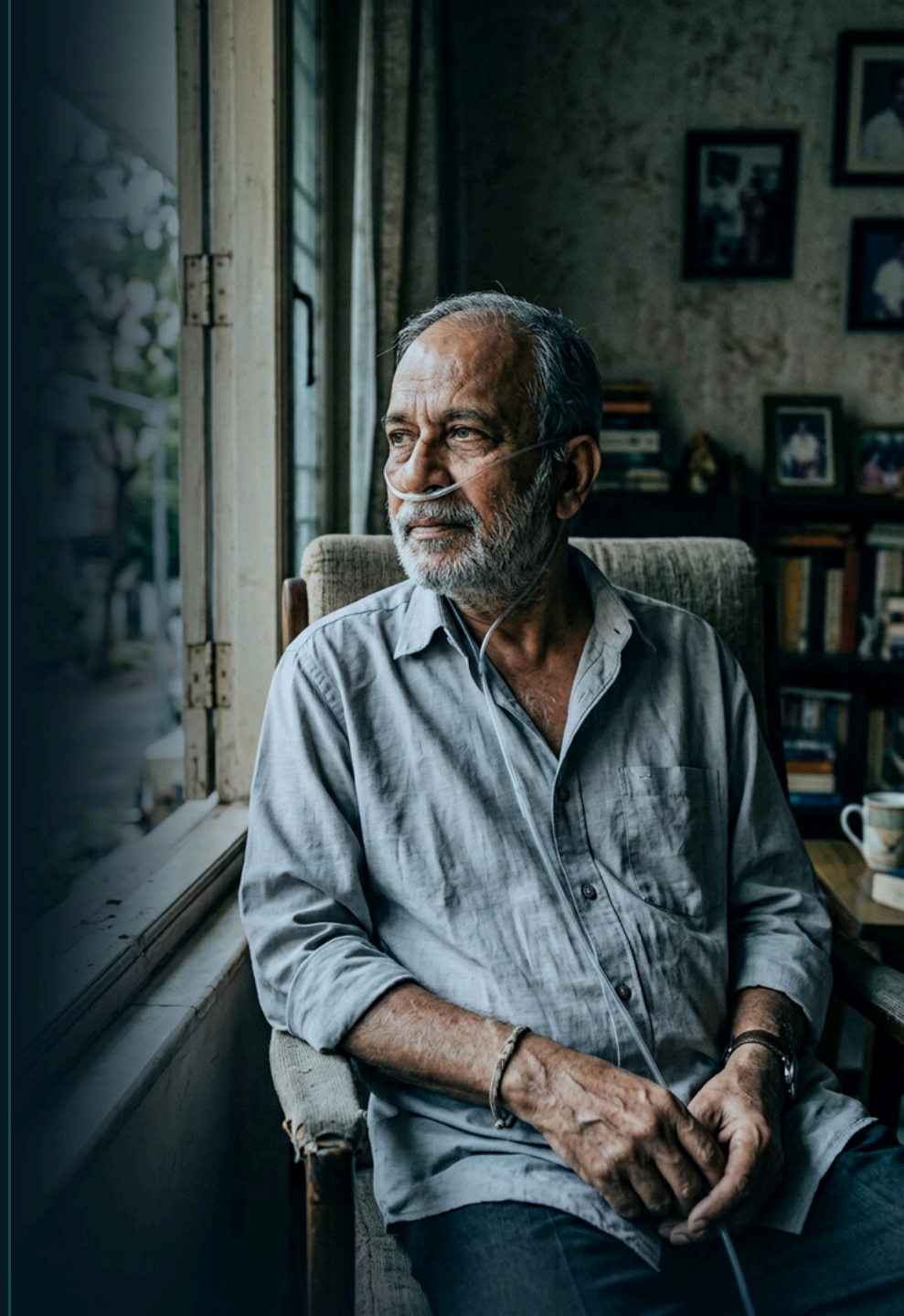
A consultation, a prescription, a hospital visit

3rd

LEADING CAUSE OF DEATH WORLDWIDE

3.4 million deaths in 2023 — about 6% of all deaths globally. COPD is not rare, and it is not curable.

Source: WHO COPD fact sheet, 10 June 2026.



ILLUSTRATIVE PHOTOGRAPH · SYNTHETIC PATIENT DATA ONLY

Care is built from snapshots. COPD is not a snapshot disease.

68% OF EXACERBATIONS GO UNREPORTED

In a longitudinal cohort, 68% of exacerbations recorded by patients in daily diaries were never reported to a health professional (327 of 486).

Langsetmo et al., AJRCCM 2008

BETWEEN TWO ROUTINE VISITS



Ninety days of lived experience. Two recorded observations.

Schematic, not to scale. Illustrative interval — not a measured statistic.

THREE POINTS ON THE PATH WHERE THE PICTURE BREAKS DOWN

01 INFORMATION IS FRAGMENTED

Results, letters, prescriptions and self-reported symptoms sit in different places and rarely join up.

02 DAILY EXPERIENCE IS RARELY CAPTURED

Symptoms are under-reported by patients — and clinicians do not systematically ask about them.

03 CHANGE IS DIFFICULT TO SEE

With a snapshot at each visit, an individual's baseline and drift between visits are hard to judge.

A SNAPSHOT IS NOT A TRAJECTORY

A single spirometry value or one saturation reading describes a moment. It does not describe how a person has been breathing since the last visit.

A SINGLE MEASUREMENT

Valid for that moment. Weak at describing direction.

A LONGITUDINAL UNDERSTANDING

Repeated records against a personal baseline.

THE RISKIEST PATIENTS CAN LOOK ORDINARY

Hurst et al., NEJM 2010: a history of exacerbations was the single best predictor of future exacerbations across all GOLD stages.

Sources: Langsetmo et al. 2008; GOLD 2026; Elliott-Button et al. 2024.

From isolated observations to a longitudinal COPD digital twin

A research prototype that keeps each patient's COPD story in one place — and labels every model output for what it is.

THE INFORMATION FLOW · CLICK TO REVEAL

01 PATIENT OBSERVATIONS
Daily wellbeing, breathlessness, sputum, adherence, physiology

↓

02 LONGITUDINAL HEALTH RECORD
A versioned twin state for every hour of the record

↓

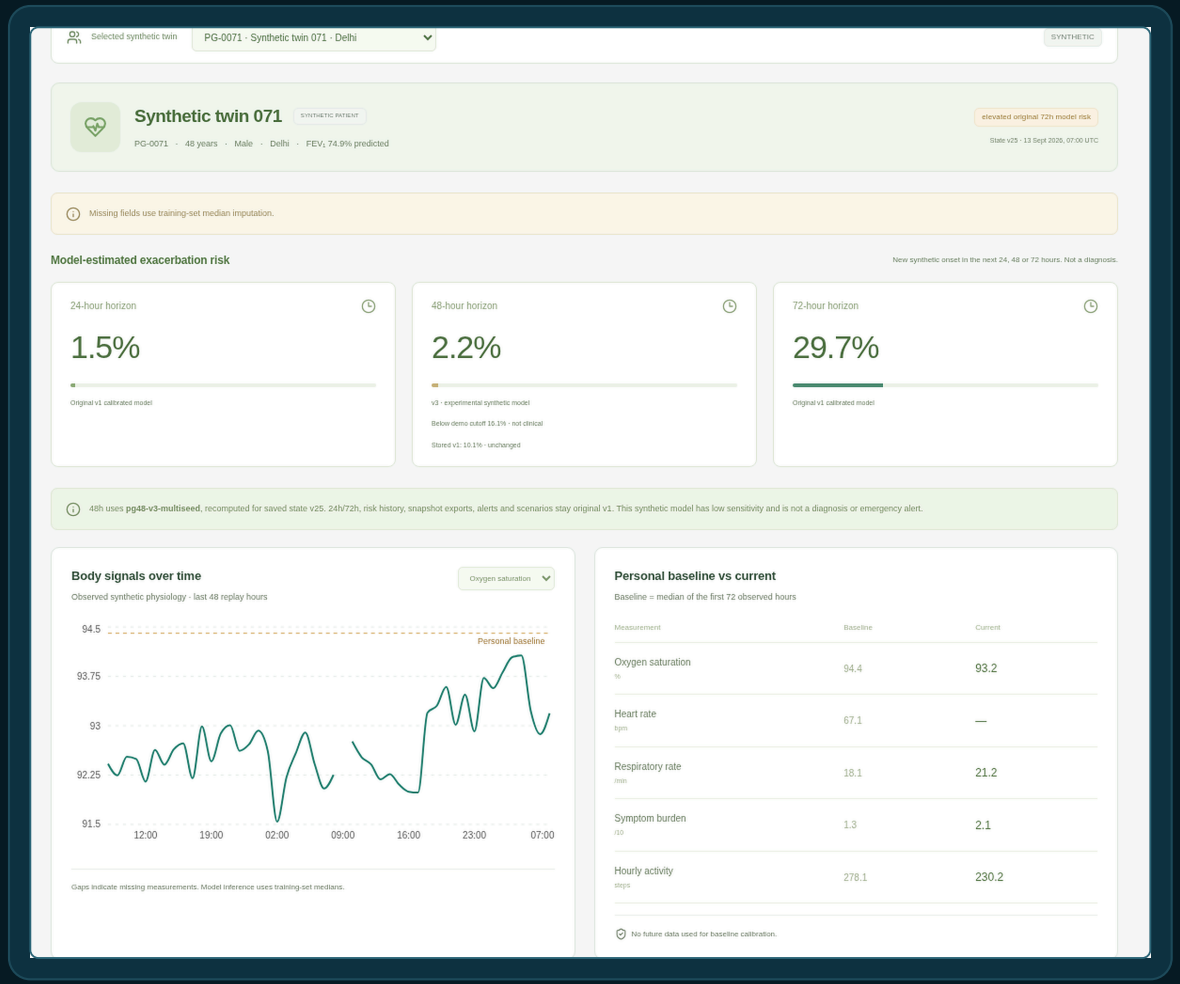
03 PERSONAL BASELINE
First 72 observed hours become that patient's own reference

↓

04 RESEARCH-MODEL INSIGHTS
Experimental 24 / 48 / 72-hour estimates, model version attached

↓

05 CLINICIAN REVIEW
A person reviews, documents and decides — the model never does



Actual PranaGuard interface: synthetic patient PG-0071, replay state v25.
Outputs shown are experimental research estimates on synthetic data — not confirmed clinical events.

Status: working software prototype · synthetic cohort of 120 patients · not a medical device · not clinically validated

What could change — if the approach is validated

Potential impact is described here as potential. The prototype has not demonstrated any of it in patients.

PATIENT IMPACT

- Better visibility of changing breathing patterns
- A structured way to record daily experience
- One continuous record instead of disconnected episodes
- Better-informed conversations with the care team

CLINICIAN IMPACT

- Consolidated longitudinal information in one view
- Change since the personal baseline, not just today
- Patient-reported information gathered and reviewed
- Research insight offered as context, never as a verdict

HEALTHCARE SYSTEM IMPACT

- A more structured model for longitudinal monitoring
- A research platform for exacerbation-associated patterns
- A foundation for future, evidence-based COPD workflows

THE IMPACT PATHWAY · POTENTIAL, NOT MEASURED

01 **Patient observations** → 02 **Structured longitudinal information** → 03 **Clinician visibility** → 04 **Informed assessment** → 05 **Continuity of care**

CURRENT EVIDENCE vs. FUTURE VALIDATION

WHAT EXISTS TODAY

- Working end-to-end software prototype, audited and reproducible
- Model experiments on a synthetic cohort of 120 patients
- Public modelled environmental context (Open-Meteo / CAMS), city-level only

WHAT MUST HAPPEN NEXT

- Clinician-supervised evaluation and prospective research
- Clinical usability and workflow assessment
- Measurement of real patient and healthcare outcomes

NOT YET DEMONSTRATED

No reduction in hospitalisations, exacerbations or mortality has been demonstrated. Model performance on synthetic data has been weak and is reported as such.

Better COPD care begins with understanding the changes happening between consultations.

01
PATIENT
daily experience

→

02
LONGITUDINAL DATA
the record over time

→

03
CLINICIAN
review and judgement

→

04
RESEARCH INSIGHTS
evidence, explicitly labelled



PranaGuard — turning fragmented COPD observations into a more connected picture of patient health.

Status today: a working research prototype on synthetic data. Clinical value is a hypothesis to be tested, not a result we are claiming.

ASK THE ROOM

What if the changes between consultations were easier to understand?

Thank you — we welcome your questions.

PranaGuard · Happiest Health Digital Twin Challenge 2026

APPENDIX · FURTHER READING

Ten answers, ready to deliver

BACKUP · NOT PART OF THE FIVE-MINUTE FLOW

WHAT IS COPD?

A common, preventable and treatable disease with persistent respiratory symptoms and airflow limitation. Symptoms include breathlessness, cough and sputum. It is not curable. (WHO; GOLD)

WHAT IS AN EXACERBATION?

An acute worsening of respiratory symptoms that leads to additional therapy. Severity ranges from extra bronchodilator use to hospitalisation. (GOLD)

WHY LONGITUDINAL MONITORING?

A single reading describes a moment. Repeated records against a personal baseline begin to describe direction of change between visits.

WHAT IS THE DIGITAL TWIN HERE?

A versioned, replayable state for each synthetic patient: observations, personal baseline, derived features and model context kept in one continuous record.

WHAT DOES THE MODEL PREDICT?

Experimental 24/48/72-hour estimates of a new synthetic exacerbation onset in the research cohort. These are research outputs, not validated probabilities or alerts.

WHAT DATA IS, AND IS NOT, REAL

Patients, physiology and outcomes are all synthetic. The only real inputs are public modelled city air-quality and weather data — context, not personal exposure. No clinical data at all.

HAS IT BEEN CLINICALLY VALIDATED?

No. There has been no clinical trial, no real patient data and no regulatory review. It is explicitly labelled a research prototype and not a medical device.

WHAT IMPACT IS DEMONSTRATED?

Software functionality only: a working prototype whose behaviour is tested and reproducible. No patient, clinical or system-level outcome has been measured.

HOW WOULD IT BE EVALUATED?

Clinician-supervised usability studies, then prospective observational research with adjudicated outcomes, then outcome measurement — under ethics approval and governance.

HOW DOES A DOCTOR USE IT?

Sign in, open an assigned patient, review the twin, trends and labelled model context, then record a review. Every action is attributed and audited.

Full source list on the next page. Every answer above is defensible from the project's own documentation or a cited publication.

APPENDIX · REFERENCES

Sources and further reading

Global burden: COPD is the third leading cause of death worldwide, with 3.4 million deaths in 2023 (about 6% of global deaths). Nearly 90% of COPD deaths under age 70 occur in low- and middle-income countries (WHO, 2026). What PranaGuard does not claim: no clinical validation, no measured outcome improvement, no regulatory status, no real patient data. All patient records in the prototype are synthetic.

REFERENCES

1. World Health Organization. Chronic obstructive pulmonary disease (COPD). Fact sheet, 10 June 2026.

2. Global Initiative for Chronic Obstructive Lung Disease. Global Strategy for the Diagnosis, Management and Prevention of COPD: 2026 Report. goldcopd.org

3. Langsetmo L, Platt RW, Ernst P, Bourbeau J. Underreporting exacerbation of chronic obstructive pulmonary disease in a longitudinal cohort. Am J Respir Crit Care Med. 2008;177(4):396-401.

4. Hurst JR, Vestbo J, Anzueto A, et al. Susceptibility to exacerbation in chronic obstructive pulmonary disease. N Engl J Med. 2010;363(12):1128-1138.

5. Elliott-Button HL, Johnson MJ, Hutchinson A, et al. “I’m living in a ‘no’ world now…” npj Primary Care Respiratory Medicine. 2024;34:37.

6. How People with COPD Perceive and Communicate Exacerbations: A Multicountry Survey Study. Int J Chron Obstruct Pulmon Dis. 2025.

7. PranaGuard project documentation: docs/METHODOLOGY.md, docs/MODEL_CARD.md, docs/model-v3/RESULTS.md, docs/LIMITATIONS.md, docs/DATA_PROVENANCE.md.

STATE OF THE EVIDENCE

IMPLEMENTED
Working prototype, audited and reproducible

EXPERIMENTAL
Synthetic-cohort model studies; weak 72-hour performance reported openly

INTENDED
Longitudinal visibility and structured record-keeping

NOT CLAIMED
Any clinical outcome, validated prediction or regulatory status

CITATION DISCIPLINE

Only the seven sources listed are used anywhere in this deck. If a jury member asks for a source that is not among them, say so plainly rather than reaching for a number you cannot defend.